



Phi

A constructed language

Daniel Ellison

April 17, 2021

1 Introduction

PHI is a constructed language. A constructed language is one whose phonology, grammar, and vocabulary, instead of having developed naturally, are consciously devised¹. The sound of the name *Phi* embodies the simplicity and softness of the language itself. *Phi* (Φ) is also beauty in mathematics. It is

the golden ratio, an irrational number with a value of approximately 1.618033988 which expresses the relationship that the sum of two quantities is to the larger quantity as the larger is to the smaller. – Wiktionary²

It is thought that the value of *phi* is one of the fundamental characteristics of nature; in the language Phi, nature is fundamental to its being. Listening to spoken Phi one is sometimes reminded of the sounds of the natural world. There is also a musicality and rhythm to sentences in the language.

¹https://en.wikipedia.org/wiki/Constructed_language

²<https://en.wiktionary.org/wiki/%CF%86>

2 Phonology

Table 1: Consonants

	Bilabial	Labial	Dental	Alveolar	Postalv.	Velar	Glottal
Plosive	p			t			
Nasal	m			n			
Trill				r			
Tap or flap				ɾ			
Fricative	ɸ	ɱ	θ	s	ʃ	ɯ	h
Approximate		w				w	
Lateral approx.				l			

Table 2: Vowels

	Front	Central	Back
Close	i		
Mid	e		ɔ
Open		ä	

3 The Tengwar

Phi can be written in either of two ways: using the standard Latin alphabet or the Tengwar script. Tengwar was invented by J. R. R. Tolkien and first described in Appendix E of the third book in his trilogy, *The Lord of the Rings*. There are numerous sources of information available concerning the Tengwar so details and history of the script will not be explored in this document.

The look and feel of the Tengwar characters (*tengwa*) and its diacritical marks (*tehta*) match the ideals and philosophies behind this language. Phi has a small set of consonants and vowels so it is relatively straightforward to map its graphemes to Tengwar characters. The Unicode font *Tengwar Telcontar*³ is used in this manual to render Phi text in Tengwar script. A custom keyboard layout⁴ was employed to facilitate typing Tengwar characters.

Tables 3 and 4 outline the mapping between the graphemes used in Phi and the corresponding Tengwar characters, as well as the keypresses necessary to produce the characters using the font and the keyboard layout mentioned previously. An application or utility that supports Graphite fonts⁵ is required for the font to render correctly. A link to details of this requirement can be found in the first footnote below.

Written Phi uses the Tengwar's Quenya mode. Vowel diacriticals are placed on the consonant that precedes the vowel. Since all Phi words start with a consonant, there is never a need to use an initial vowel carrier. Words in Phi often contain two consecutive vowels, but a vowel carrier is still unnecessary: the second vowel is placed below the same consonant. The letters would then be read as consonant, upper vowel, then lower vowel. This and the use of a single Tengwar character for consonant doubles such as ph or sh cause Phi script to be very compact and efficient.

³<http://freetengwar.sourceforge.net/tengtelc.html>

⁴<http://freetengwar.sourceforge.net/keylayouts.html>

⁵https://scripts.sil.org/cms/scripts/page.php?site_id=projects&item_id=graphite_fonts

The Phi word for “love” is *lothea*. In the Tengwar script it looks like this:



This six-letter word in latin characters requires only two *tengwa* to represent it. Many words in Phi are two to three letters long. This occurs mainly in parts of speech such as particles, prepositions, or pronouns. These words are almost always a single *tengwa* with diacriticals in Tengwar script. All of this results in a very concise writing system.

This is the phrase “I love you” in its simplest vernacular form in Phi:



That is a beautiful sentiment rendered in a beautiful script. In more formal communication in Phi, particles are used to mark parts of speech in order to reduce ambiguity. As well, word markers are often used to distinguish multisyllabic words for clarity. The Phi Tengwar punctuation is outlined in Table 5. As an example, the phrase “I love you” would be written formally as:



Table 3: *Phi consonant grapheme to Tengwar mapping*

Grapheme	Tengwa	Keypress
p	Ṗ	p
t	ṭ	t
m	Ṣ	m
n	ṣ	n
r	Ṛ	shift-r
s	Ṡ	s
ś	ṡ	shift-s
h	Ḥ	h
w	Ṛ	w
l	Ṭ	l
ph	Ḃ	shift-p
th	ḥ	shift-t
sh	Ḍ	shift-c
wh	Ṭ	shift-k

Table 4: *Phi vowel grapheme to Tengwar mapping*

Grapheme	Tehta	Keypress
i		i / option-i ⁶
e		e / option-e
o		o / option-o
a		a / option-a

Table 5: *Phi punctuation to Tengwar mapping*

Punctuation	Tengwar	Keypress
space	·	,
period	:	shift-,

⁶ Keypresses for the low diacriticals are given for macOS. Use *ctrl-alt* on Windows instead of *option*.

4 Examples

Following are several examples of Phi phrases in formal mode. They are given in romanized text and English, followed by the Tengwar script.

si phiathe soa mia na thanie te phi – My name is Daniel.

ဇ.နု.ဇ.မု.ကံ.နံ.ဂံ.ပ

si mia na thi te lothea – I love you.

ሪ.፱.፳.ከ.፶.ረከ

si whetho na mipho te phi – The sky is blue.

ሪብራክ፡ጳውሎስ፡

si lo sha na noshea te lephe – They like food.

ᆞ.ᆢ.ᆣ.ᆤ.ᆥ.ᆦ.ᆧ

si mia na sha te lephe – I like it.

ᚠ.ᚹ.ᚳ.ᚩ.ᚱ.ᚼ

ho si mia na hashe te me lephe – I do not like green!

X·Z·Ṁ·Ṛ·X̣d·ṗ·m̄·čḂ

*si mia na miw*ho soa *mia te li lep*he – I liked my friend.

ሪ.ጠ.ክ.ጠደ.ቫ.ጠ.የ.ፒ.ፔ.ፕ

si noshea na soa lo mia te phi – The food is ours.

လက်ထပ်သောအခါ

si noshea soa lo mia na shishoa te phi – Our food is good.

ሪ.ጠፋ.ቫ.ረ.ጃ.ጠ.ረፋ.ፆ.ኔ

wha si thi na ropo shishoa te la phe – Are you in good health?

፳.ሪ.ክ.ጠ.ፍኔ.ረፋ.ፆ.ረ.ኔ

si lo mia na lo lipha sathe te lothea – We love the orange trees.

ሪ.ረ.ጃ.ጠ.ረ.ረ.ኔ.ፃክ.ፆ.ረክ